EXPERIMENT – 1

ARDUINO UNO-BASED SMART HOME SAFETY AND OCCUPANCY MONITORING SYSTEM USING GAS, PIR AND ULTRASONIC SENSORS

LABORATORY PERFORMANCE ASSESSMENT RUBRICS

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Develop the Proteus circuit	Circuit Design and Configuration	Independently develops the complete multi-sensor circuit with correct connections	Develops the circuit with minor connection errors	Develops a basic circuit with considerable assistance	Unable to develop the required circuit correctly	25
Develop the Embedded C program	Program Development	Writes and organizes the program correctly with proper sensor acquisition and control logic	Develops the program with minor errors	Develops a basic program with significant assistance	Unable to develop the required program	25
Implement sensor processing	Sensor Acquisition and Decision Logic	Correctly implements gas, PIR and ultrasonic processing and threshold decisions	Implements the sensor logic with minor errors	Implements basic sensor processing with assistance	Unable to implement the required logic	25
Verify system operation	Simulation and Result Analysis	Correctly verifies and clearly explains all sensor outputs and alert conditions	Verifies operation with minor interpretation gaps	Obtains partial output with limited explanation	Unable to obtain or interpret the output	25
Total						**100**

EXPERIMENT – 2

FORMAL DESIGN OF AN EMBEDDED CONTROLLER USING STATE MACHINE (STATECHART) MODELING – TRAFFIC LIGHT CONTROLLER

LABORATORY PERFORMANCE ASSESSMENT RUBRICS

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Develop the FSM/Statechart model	State Modeling	Independently develops correct states, transitions, timing conditions and outputs	Develops the model with minor errors	Develops a basic model with considerable assistance	Unable to develop the required FSM model	25
Develop the Proteus circuit	Circuit Design and Configuration	Independently develops the complete circuit with correct Arduino, LEDs, resistors and Virtual Terminal connections	Develops the circuit with minor connection errors	Develops a basic circuit with significant assistance	Unable to develop the required circuit	25
Develop the Embedded C program	Program Development	Correctly implements the FSM with appropriate state transitions and outputs	Implements the FSM with minor errors	Develops a basic program with significant assistance	Unable to implement the required program	25
Verify the simulation	Simulation and Result Analysis	Correctly verifies all states and clearly explains the state transitions and outputs	Verifies operation with minor interpretation gaps	Obtains partial output with limited explanation	Unable to obtain or interpret the output	25
Total						**100**

EXPERIMENT – 3

I2C-BASED HEALTH MONITORING NODE FOR PULSE AND TEMPERATURE ACQUISITION

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Develop the Proteus circuit	Circuit Design and Configuration	Independently develops the complete circuit with correct Arduino, LM75AD, I²C and pulse connections	Develops the circuit with minor connection errors	Develops a basic circuit with considerable assistance	Unable to develop the required circuit correctly	25
Develop the Embedded C program	Program Development	Writes and organizes the program correctly with proper I²C, pulse-counting and serial operations	Develops the program with minor errors	Develops a basic program with significant assistance	Unable to develop the required program	25
Implement temperature and pulse acquisition	Sensor and Signal Processing	Correctly acquires temperature and calculates BPM	Implements acquisition with minor errors	Obtains partial output with assistance	Unable to implement the required acquisition	25
Verify the health monitoring node	Simulation and Result Analysis	Correctly verifies and clearly explains temperature and BPM outputs	Verifies operation with minor interpretation gaps	Obtains partial output with limited explanation	Unable to obtain or interpret the output	25
Total						**100**